South Korea: energy security, economic opportunity and partnership priorities

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AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- South Korea is most exposed in upstream gas, upstream oil and upstream nuclear, where its Energy Security is higher than only 0%, 1% and 7% of the other economies scored. In upstream oil and gas, production falls short of consumption and no reserves are recorded.[1]
- Its strongest openings are in downstream green hydrogen, downstream gas and downstream nuclear, where its Economic Opportunity is higher than 97%, 96% and 94% of the other economies scored. Use per person, production and investment lift them.[1]
- On the paper's weighting, the Strategic Index ranks downstream batteries 1st and midstream batteries 2nd among its own sectors. Yet 5 sectors combine top priority with the least policy, including upstream nuclear.[1]
- The Partnership Strength Index rates the United States highest for South Korea, as a reliable supplier. China, its 1st trade partner, ranks 35th on Partnership Strength.[1,2]

1. Energy security risks and why

6 of South Korea's 29 placed sectors, those whose scores can be compared with other economies, sit in the two low-energy-security corners. Energy Security measures how well a country can rely on what it needs in a sector. Measured against other economies, its lowest Energy Security is in upstream gas (0.20), upstream oil (0.35) and upstream nuclear (0.16). These are higher than only 0%, 1% and 7% of the other 97, 96 and 174 economies scored. The upstream oil and gas sectors sit in the Secure corner and upstream nuclear in the Upgrade corner. The paper's framework associates the Secure corner with securing supply through trusted partners and the Upgrade corner with building domestic capability.[1]

In upstream gas and upstream oil, energy imports and reserves pull the scores down. Gas production minus consumption was -2.27 EJ in 2025, and for oil it was -5.42 EJ. Both take production from the U.S. Energy Information Administration's international statistics, as the Energy Institute does not report it separately. No proved gas or oil reserves are recorded, which the index counts as zero; reserves carry 28.6% of the weight of each score. For context, its largest suppliers of upstream gas goods in 2024 were the United States (21.8% of imports, which totalled USD 34.7 bn), Australia (20.8%) and Qatar (19.1%).[1,2,3,4]

Its high-priority sectors, those in the Upgrade corner ranked by the Strategic Index, are upstream nuclear and downstream oil. In upstream nuclear, the score is low mainly because of trade. On its trade deficit in these goods relative to gross domestic product, South Korea scores higher than only 6% of the other economies. In downstream oil, consumption pulls the score down: on projected growth in use per person, South Korea scores higher than only 2% of the other economies.[1]

2. Economic opportunities and why

18 of the 29 placed sectors sit in the two high-opportunity corners, where Economic Opportunity is above the sector's middle economy. Economic Opportunity measures how well placed a country is to create value in a sector. The highest Economic Opportunity scores are in downstream green hydrogen, downstream gas and downstream nuclear, all in the Grow corner. They are higher than 97%, 96% and 94% of the other 157, 98 and 66 economies scored. The paper's framework associates the Grow corner with export promotion.[1]

In downstream green hydrogen and downstream gas, consumption pushes the score up: on use per person today South Korea scores higher than 99% and 99% of the other economies. Production also adds to the score in downstream gas and downstream nuclear, where generation was 174 TWh and 185 TWh in 2025. In downstream nuclear, investment pushes the score up as well.[1,3]

Among midstream sectors, midstream electric vehicles stands out, higher than 93% of the other economies scored. Investment and trade push it up: South Korea reported exports of USD 41.1 bn of these goods in 2024, and its share of world exports was 4.5% in 2023. For context, on the reconciled trade data, the United States took 38.6% of its exports of these goods in 2024.[1,2,5,6]

3. Sectoral priorities and the existing policy posture

The Strategic Index ranks a country's own sectors by combining opportunity, security need, policy activity, the stage and the technology's low-emission value. On the paper's weighting, it puts downstream batteries 1st (0.66), midstream batteries 2nd and upstream solar 3rd among South Korea's own sectors, each 0.020, 0.020 and 0.015 ahead of the next. Then come upstream batteries (4th) and upstream green hydrogen (5th), which are almost tied, 0.002 apart.[1]

The allied pick, its leading upstream or midstream sector once sectors without a recorded resource base are set aside, is midstream batteries, 0.035 ahead of upstream batteries. Ranked above the runner-up, upstream solar is set aside: no resource base is recorded for it.[7]

Policy activity counts International Energy Agency Policies and Measures records and the Climate Action Tracker rating, read against each technology. Priority here means the Strategic Index without its policy term, unlike the Upgrade-corner sense above. The sector with top priority and the most policy is midstream batteries. By contrast, upstream nuclear, upstream wind, upstream green hydrogen, upstream electric grid and midstream electric grid combine top priority with the least policy. South Korea has 0 records naming both the technology and the stage of upstream nuclear. The reverse holds for downstream gas, downstream oil, downstream green hydrogen, downstream electric vehicles and downstream coal, which combine bottom priority with the most policy. South Korea's Climate Action Tracker rating is Highly insufficient (July 2023), and it has 247 Policies and Measures records. For context, Country Insights also lists industrial-policy measures from Global Trade Alert; policy activity does not count them.[1,8,9]

4. Partnership priorities and why

The Partnership Strength Index rates partners as suppliers (Friendshore), as markets for exports (Partnership Opportunity) and, for middle- and lower-income partners, by Development Potential. From South Korea's viewpoint it ranks the United States 1st, Australia 2nd, the United Kingdom 3rd, Canada 4th and the Netherlands 5th of 193 partners. All but the United Kingdom, an export market, stand out most as a reliable supplier. As a supplier, the United States scores best in upstream gas, upstream oil, midstream

batteries, downstream geothermal and upstream nuclear. These include upstream oil, gas and nuclear, where South Korea's Energy Security is low.[1]

Some of these priorities match today's ties. The United States is South Korea's 2nd trade partner on average across its energy-goods baskets, Australia its 4th and Canada its 7th. The United States and South Korea both belong to the Indo-Pacific Economic Framework, the Minerals Security Partnership and a U.S. free-trade agreement. The United Kingdom and the Netherlands rank 22nd and 20th as trade partners today. Conversely, China is its 1st trade partner, with 28.3% of its imports on average, but ranks 35th on Partnership Strength. UN votes put South Korea and China in opposite blocs.[1,2,10,11]

As a partner to the coalition, South Korea ranks 14th of 192 economies, on Partnership Strength averaged over the core members' viewpoints. In the allied network design, which spreads upstream and midstream production across coalition members, its largest designed share is 18.6% of midstream electric vehicles, followed by 14.5% of midstream oil.[1,7]

5. Data notes and caveats

Baseline of 12 October 2026; details in Country Insights.

- Scores compare South Korea with other economies in the same sector; they are not measures of risk or profit in absolute terms.
- All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings.
- Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order.
- Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up.
- Some figures behind these scores are filled in where the source has no value: with zero where a source lists every holder, otherwise with the world average.
- Some of South Korea's sectors are not placed: a score is compared with other economies only when the economy is sovereign and at most half of the score rests on filled-in data.

Sources

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Allied Industrial Base Explorer, Country Insights: South Korea